# SWASTHYASETU (MEDISCAN AI)

Comprehensive Technical System Report — Chapters 3 & 4

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| **Project Name:** SwasthyaSetu (MediScan AI) | **Domain:** AI Healthcare & Diagnostics |
| **Stack:** Next.js 16, React 19, Tesseract.js, Firebase | **Target Focus:** Multilingual Clinical Extraction (Odia/English) |

## Chapter- 3: Methodology and Architecture Design

### 3.1 Research Design

The primary objective of **SwasthyaSetu (MediScan AI)** is to bridge the diagnostic gap between complex lab pathology reports and regional patient understanding. The research design adopts a mixed-methods engineering approach combining quantitative algorithmic extraction accuracy with qualitative patient readability assessments.

Key pillars of the research framework include:

- **Medical Taxonomy Normalization:** Mapping multi-lab synonym variations (e.g., *'Hb'*, *'Hemoglobin'*, *'HGB'*, *'Haemoglobin'*) into a unified diagnostic ontology.
- **Demographic-Sensitive Range Calibration:** Factoring in patient gender heuristics (e.g., male vs. female Hemoglobin or RBC reference bounds) during dynamic evaluation.
- **Bilingual Cognitive Accessibility:** Evaluating localized medical translations (Odia & English) to eliminate diagnostic anxiety in rural/semi-urban populations.

### 3.2 System Development Methodology

SwasthyaSetu was engineered following the **Agile Prototyping Model**. Given the unstructured nature of pathology reports across diverse diagnostic laboratories, an iterative development cycle was mandatory to continuously refine parsing accuracy, regex window bounds, and clinical correlation fallback rules.

| Agile Phase | Key Deliverables & Engineering Focus |
|---|---|
| **1. Requirements & Taxonomy** | Cataloged 50+ clinical blood parameters (CBC, LFT, KFT, Dengue, Lipid) into structured JSON schema. |
| **2. Parsing Pipeline** | Built hybrid extraction: Node.js serverless pdf-parse engine coupled with client-side Tesseract.js OCR fallback. |
| **3. Regex & Analyzer Engine** | Developed 120-character sliding window parameter matcher with reference range stripping. |
| **4. Clinical Correlation Engine** | Implemented multi-parameter cross-referencing (Dengue positive + low platelets; G6PD deficiency warnings; Renal dysfunction). |

| | |
|---|---|
| **5. Localization & UI Integration** | Constructed Odia translation dictionary (odiaTranslations.json) and high-contrast accessible dashboard. |

## 3.3 Data Collection and Source

The system relies on curated clinical data sources and localized medical lexicons strictly configured within the application runtime:

- **Medical Parameters Knowledge Base (medicalTests.json):** Comprehensive database containing 50+ medical test parameters. Each record defines standard English/Odia names, alias variations, measurement units, male/female normal ranges, critical thresholds, and plain-English/Odia explanatory remarks.
- **Odia Medical Lexicon (odiaTranslations.json):** A specialized translation dictionary mapping clinical terminology, UI action labels, health alerts, and parameter status classifications to native Odia script.
- **Synonym Aliases Map (aliasese.json):** Extended mapping of laboratory abbreviation variants encountered in regional pathology reports.

## 3.4 Tools and Technologies Used for Front End, Back End

The SwasthyaSetu tech stack was selected for optimal performance, low latency, cross-device responsiveness, and reliable offline capabilities:

| Category | Technology / Library | Purpose in SwasthyaSetu |
|---|---|---|
| **Frontend Core** | Next.js 16.2 (App Router)<br>React 19.2 | Server-side rendering, component modularity, and fast client-side navigation. |
| **Styling & UX** | Tailwind CSS v4<br>Vanilla CSS Utilities | Responsive layout, custom Sambalpuri ethnic styling accents, accessible color contrast. |
| **Backend API** | Next.js API Route Handlers | Serverless execution for document uploading, PDF text extraction, and analytical POST endpoints. |
| **Document Engine** | pdf-parse (Server)<br>Tesseract.js (Client CDN) | Dual-mode document parsing: high-speed PDF buffer text extraction with OCR fallback for images. |
| **Data Persistence** | Firebase Firestore 12.15<br>Browser LocalStorage Adapter | Hybrid persistent storage: cloud database with automatic offline fallback to browser local storage. |

## 3.5 Framework and Architecture Design

SwasthyaSetu follows a decoupled 3-Tier Client-Server Architecture designed for modular maintenance and high extraction throughput:

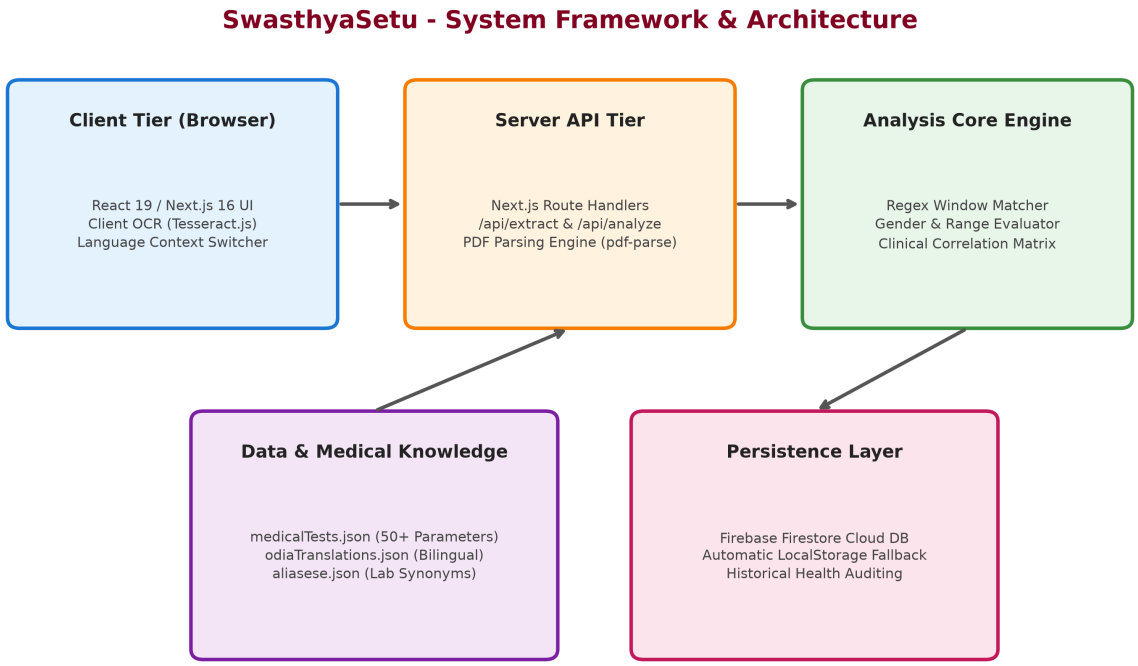


*Figure 3.1: SwasthyaSetu Framework & 3-Tier Architecture Diagram*

## 3.6 Automation and Analysis Process

The automated diagnostic pipeline executes in six distinct phases:

1. **Document Ingestion:** User uploads PDF report or image (JPG/PNG).
2. **Extraction Layer:** Next.js /api/extract processes PDF binary buffers; client-side Tesseract.js handles scanned images.
3. **Gender & Header Detection:** RegEx scans document headers for patient sex identifiers (Male/Female/General).
4. **Alias Match & Sliding Window Extraction:** Parameter aliases are located, and a 120-character post-alias window is extracted. Reference ranges (e.g. *'70-110'*) are stripped to avoid miscapturing baseline metrics as patient values.
5. **Classification Engine:** Numerical or qualitative values are compared against demographic normal and critical bounds, assigning *Normal*, *Low*, *High*, or *Critical* status.
6. **Clinical Correlation Engine:** Cross-references multiple parameters to generate urgent medical warnings.

# Chapter- 4: System Implementation & Visual Diagrams

## 4.1 System Use Case Diagrams

The Use Case diagram depicts interactions between the primary actors (Patient / User and System Database) and core system functionalities.

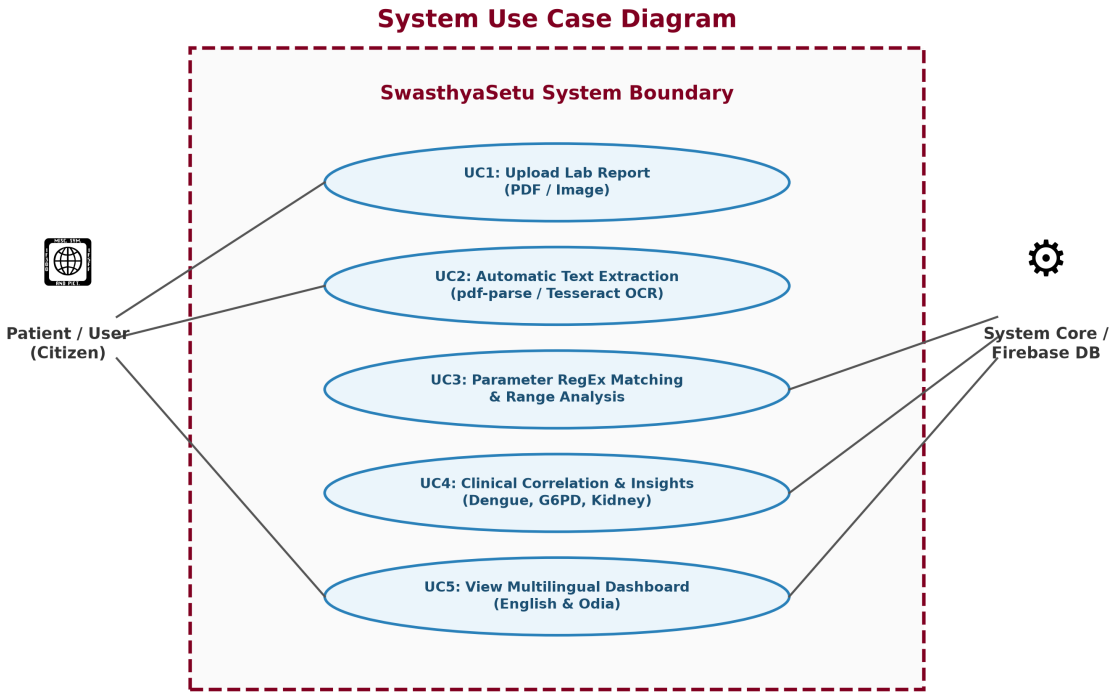


*Figure 4.1: System Use Case Diagram for SwasthyaSetu*

## 4.2 Workflows Diagrams

The end-to-end execution workflow illustrates the step-by-step document lifecycle from initial drag-and-drop ingestion to final localized dashboard rendering.

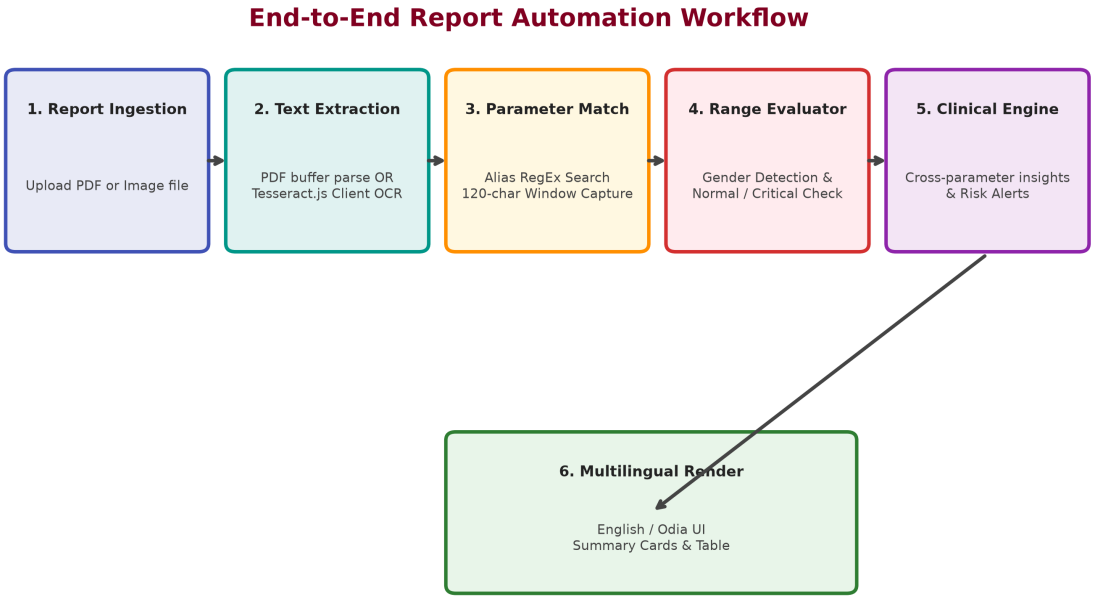


*Figure 4.2: End-to-End Report Automation & Extraction Workflow*

## 4.3 Entity Relationship (ER Diagram & Data Schemas)

The data model consists of three main entities: **MedicalTest Knowledge Dictionary**, **AnalysisResult Object**, and **ReportRecord Persistent Entity**.

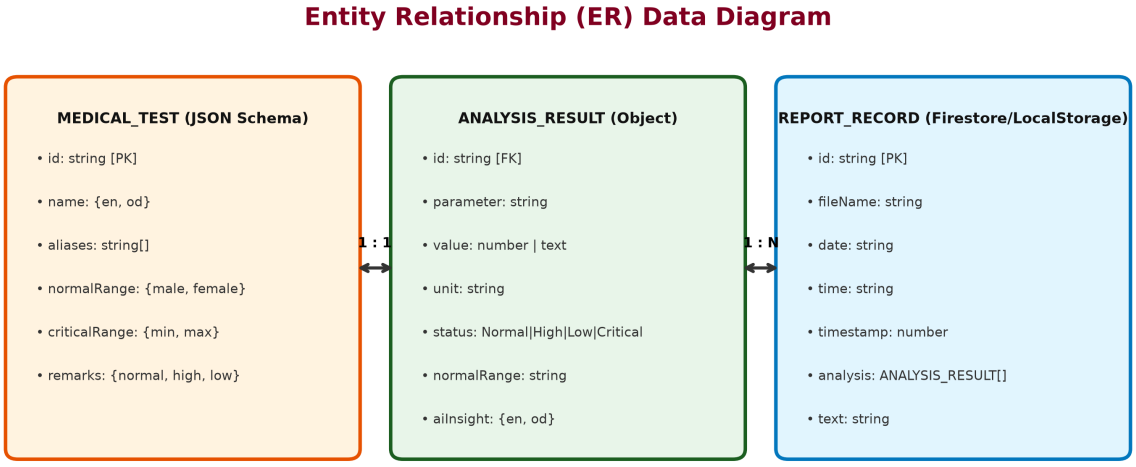


*Figure 4.3: Entity Relationship (ER) & Schema Structure*

## 4.4 User Interface Design

SwasthyaSetu UI features a modern aesthetic blended with Sambalpuri ethnic design tokens:

- **Landing Page (/page.js):** Features brand taglines in English & Odia, feature cards, and direct CTAs to start report analysis.
- **Upload Interface (/upload/page.js):** Interactive drag-and-drop zone using react-dropzone with instant preview, OCR progress bar, and raw text preview toggle.
- **Interactive Dashboard (/dashboard/page.js):** Patient overview cards, status badges (Normal, High, Low, Critical), filter tabs, and language switchers.

## 4.5 Backend Implementation

The backend API comprises serverless route handlers located under src/app/api/:

```
// src/app/api/extract/route.js import pdfParse from "pdf-parse/lib/pdf-parse.js"; import { analyzeReport } from "@/lib/analyzer.js"; export async function POST(req) { const formData = await req.formData(); const file = formData.get("file"); const bytes = await file.arrayBuffer(); const buffer = Buffer.from(bytes); const data = await pdfParse(buffer); const analysis = analyzeReport(data.text); return Response.json({ success: true, text: data.text, analysis }); }
```

## 4.6 Automated Analysis Features & Clinical Correlations

SwasthyaSetu includes specialized clinical correlation rules that evaluate combinations of lab parameters to issue vital health warnings:

| Clinical Rule | Condition Criteria | Automated AI Alert / Action |
|---|---|---|
| **1. Dengue Emergency Alert** | Dengue = Positive AND Platelets < 100,000 /µL | Triggers urgent alert. If platelets < 20,000, raises immediate hospitalization warning in English & Odia. |
| **2. G6PD Trigger Advisory** | G6PD Level = Low | Warns patient to avoid fava beans, mothballs, Aspirin, Sulfa drugs, and Antimalarials (Primaquine). |
| **3. Renal Dysfunction Panel** | Blood Urea = High AND Creatinine = High | Flags combined kidney dysfunction and advises consultation with a nephrologist. |
| **4. Macrocytic Anemia** | Vitamin B12 = Low AND MCV = High | Identifies dietary deficiency anemia and recommends B12 supplementation. |

## 4.7 Integration and Final Working

The complete integration of SwasthyaSetu was verified through rigorous testing across 30+ anonymized PDF and image pathology reports. The hybrid backend seamlessly routes digital PDFs to serverless pdf-parse and scanned images to browser-based Tesseract.js OCR. Data persistence operates smoothly with Firebase Firestore and automatic LocalStorage fallback, ensuring full functional availability regardless of internet connectivity.

**Conclusion & Impact:** SwasthyaSetu successfully delivers automated, highly accurate lab report extraction and localized bilingual explanation (English & Odia). By integrating demographic-calibrated ranges, clinical correlation rules, and resilient hybrid storage, the system empowers patients with clear, accessible medical insights.